ANSWER KEY — Integration Lesson 25 Test: Inverse Trig Antiderivatives

For the grader · full working shown · give credit for correct method with small slips · a missing + C on an indefinite integral costs the point only if it happens repeatedly

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1. Evaluate exactly: (a) arctan 1 (b) arcsin(1/2) (c) arctan 0.

> **Work:** read the trig table backward. (a) $\tan(\pi/4) = 1 \rightarrow \arctan 1 = \pi/4$. (b) $\sin(\pi/6) = 1/2 \rightarrow \arcsin(1/2) = \pi/6$. (c) $\tan 0 = 0 \rightarrow \arctan 0 = 0$. **Answer: (a) π/4 (b) π/6 (c) 0.** (Grader: answers must be in radians; 45° etc. gets half credit at most.)

2. Find $\int 1/(1+x^2)\,dx$.

> **Work:** this is the boxed arctan fact — power rule can't apply (bottom is a sum), u-sub has no $du = 2x\,dx$ available. **Answer: arctan x + C.** (Grader: $\ln(1+x^2) + C$ is the classic wrong answer — its derivative is $2x/(1+x^2)$, not $1/(1+x^2)$.)

3. Find $\int 3/\sqrt{1-x^2}\,dx$.

> **Work:** constant-multiple rule + the arcsin fact: $3 \int 1/\sqrt{1-x^2}\,dx = 3 \arcsin x$. Check: $(3 \arcsin x)' = 3/\sqrt{1-x^2}$ ✓. **Answer: 3 arcsin x + C.**

4. Compute exactly: $\int_1^{\sqrt{3}} 1/(1+x^2)\,dx$.

> **Work:** $[\arctan x]_1^{\sqrt{3}} = \arctan \sqrt{3} - \arctan 1 = \pi/3 - \pi/4 = 4\pi/12 - 3\pi/12 = \pi/12$. **Answer: π/12.** (Grader: subtracting in the wrong order gives $-\pi/12$ — a negative area for a positive function should have triggered a sanity alarm.)

5. Compute exactly: $\int_0^{\sqrt{3}/2} 1/\sqrt{1-x^2}\,dx$.

> **Work:** $[\arcsin x]_0^{\sqrt{3}/2} = \arcsin(\sqrt{3}/2) - \arcsin 0 = \pi/3 - 0$. **Answer: π/3.**

6. Find $\int 1/(16+x^2)\,dx$.

> **Work:** $a^2 = 16 \rightarrow a = 4$; a-form gives $(1/a)\arctan(x/a)$: $(1/4)\arctan(x/4) + C$. Check: derivative $= (1/4)\cdot 1/(1+x^2/16)\cdot(1/4) = (1/16)\cdot 16/(16+x^2) = 1/(16+x^2)$ ✓. **Answer: (1/4) arctan(x/4) + C.** (Grader: forgetting the front 1/4 is the #1 slip — take the point.)

7. Compute exactly: $\int_0^{\sqrt{2}} 1/\sqrt{4-x^2}\,dx$.

> **Work:** $a = 2$, antiderivative $\arcsin(x/2)$ (no front factor for arcsin). $[\arcsin(x/2)]_0^{\sqrt{2}} = \arcsin(\sqrt{2}/2) - \arcsin 0 = \pi/4 - 0$. **Answer: π/4.**

8. Three lookalikes: (a) $\int x/(1+x^2)\,dx$ (b) $\int 1/(1+x)^2\,dx$ (c) $\int 1/(1+x^2)\,dx$.

> **Work:** (a) x on top ≈ derivative of bottom → u-sub → ln: $u = 1+x^2$, $x\,dx = du/2$ → $(1/2)\ln(1+x^2) + C$. (b) square on the parenthesis → power rule: $u = 1+x$, $\int u^{-2}du = -1/u$ → $-1/(1+x) + C$. (c) plain 1 on top, square on x alone → arctan fact → $\arctan x + C$. **Answer: (a) u-sub→ln, $(1/2)\ln(1+x^2)+C$; (b) power rule, $-1/(1+x)+C$; (c) arctan, $\arctan x + C$.** (Grader: ask for BOTH tool name and antiderivative; 2 of 3 fully right = half point.)

9. Verify that $(1/3)\arctan(x/3)$ is an antiderivative of $1/(9+x^2)$.

> **Work:** $d/dx[(1/3)\arctan(x/3)] = (1/3) \cdot 1/(1+(x/3)^2) \cdot (1/3)$ [chain rule: inside $x/3$ has derivative $1/3$] $= (1/9) \cdot 1/(1+x^2/9) = (1/9) \cdot 9/(9+x^2) = 1/(9+x^2)$ ✓. **Answer: derivative equals $1/(9+x^2)$, verified.** (Grader: the chain-rule factor 1/3 must appear explicitly; a "verification" that never multiplies by 1/3 earns no credit.)

10. Show $\int_0^1 8/(1+x^2)\,dx = 2\pi$ and explain the connection to computing digits of π.

> **Work:** $8 \int_0^1 1/(1+x^2)\,dx = 8[\arctan x]_0^1 = 8(\pi/4 - 0) = 2\pi$ ✓. Explanation: the integral is an area under a concrete curve, so a numerical method (like the trapezoidal rule) can approximate it to any accuracy WITHOUT knowing π — and since the exact value is 2π, halving the estimate manufactures π's digits from areas alone. **Answer: $8 \cdot \pi/4 = 2\pi$; approximating the same area numerically yields digits of π.**